

Attempt the following

1. Write a Java program to create a thread that prints its name and "Hello, World!" by extending Thread class.
2. Write a Java program to create a thread that prints its name and "Hello, World!" by implementing Runnable interface.
3. Write a Java program to create a number of threads that each thread prints its name and "Hello, World!" using executor.
4. Write a Java program to get the state of the currently executing thread
5. Write a Java program to demonstrate the Thread.join() method
6. Write a Java program to get information of the current executing thread
7. Write a Java program to demonstrate the Thread priority.
8. Write a Java program that creates two threads to find and print even and odd numbers from 1 to 20.
9. the following program prints the three sentences shown below on screen in any order by three threads. The threads print one word at a time.

"Unity university is the best in Ethiopia";

"Threads are lightweight processes";

"Generics enable non-primitive data types to be parameters "

run the program repeatedly

A. Does it print the desired output? Why?

B. How can it be corrected? explain

```
1 class StringPrinter{
2     void printString(String [] str){
3         for(String s: str)
4             System.out.print(s + " ");
5         System.out.println();
6     }
7 class MyThread implements Runnable{
8     StringPrinter sp;
9     String[] str;
10    MyThread(StringPrinter sp, String[] str){
11        this.str = str;
12        this.sp = sp;
13    }
14    public void run(){
15        sp.printString(str);
16    }
17 public class ThreadExercise {
18     public static void main(String[] args) {
19         StringPrinter sp = new StringPrinter();
20         String[] s1 = {"Unity", "university", "is", "the", "best", "in", "Ethiopia"};
21         String[] s2 = {"Threads", "are", "lightweight", "processes"};
22         String[] s3 = {"Generics", "enable", "non-primitive", "data", "types", "to",
23                        "be", "parameters "};
24         ExecutorService es = Executors.newCachedThreadPool();
25         es.execute(new MyThread(sp, s1));
26         es.execute(new MyThread(sp, s2));
27         es.execute(new MyThread(sp, s3));
28     } }
```

10. Rewrite the run method of the MYThread class as follows

```
public void run(){
    synchronized(sp){
        sp.printString(str);
    }
}
```

Does it print the desired output? Why?

11. Write a Java program to create and start multiple threads that increment a shared counter variable concurrently.
12. The producer thread of the following program writes 1 to 10 to an int array of size 4, while the consumer thread reads them concurrently. Run the program repeatedly.
- A) Does it work correctly? Why? How do you correct it?
- B) Rewrite the program using ArrayBlockingQueue class of java.util.concurrent package

```
class MyBuffer{
    int[] arr = new int[4];
    int readIndex = 0;
    int writeIndex = 0;
    int count = 0;
    void produce(int value)throws Exception{
        arr[writeIndex] = value;
        System.out.println("producer writes: " + value + " at "+writeIndex);
        writeIndex = ++writeIndex%4;
        count++;
    }
    void consume() throws Exception{
        System.out.println("consumer reads: " + arr[readIndex] + " from "+readIndex);
        readIndex = ++readIndex%4;
        count--;
    }
}

class Producer implements Runnable{
    MyBuffer buffer;
    Producer(MyBuffer b){
        buffer = b;
    }
    public void run(){
        try{
            for(int i = 1; i<=10;i++)
                buffer.produce(i);
        }catch(Exception e){}
    }
}

class Consumer implements Runnable{
    MyBuffer buffer;
    Consumer(MyBuffer b){
        buffer = b;
    }
    public void run(){
        try{
            for(int i = 1; i<=10;i++)
                buffer.consume();
        }catch(Exception e){}
    }
}
```

```
public class Communication {  
    public static void main(String args[]) {  
        ExecutorService es = Executors.newCachedThreadPool();  
        MyBuffer buffer = new MyBuffer();  
        es.execute(new Producer(buffer));  
        es.execute(new Consumer(buffer));  
    }  
}
```

13. what is the difference between synchronized collections and concurrent collections?